

Assignment 3

1) How do you use the sumif function to add values that meet a specific criterion?

A) Sumif :-

The sumif function adds the values in a range only if they meet a condition.

Syntax:-

sumif(range, criteria, sum-range)

range → cells to check for the condition.

criteria → the condition

sum-range → cells to add (if the condition is true)

Ex:-

A	B
Apple	40
Mango	60
Apple	20

Q) Add the amounts for Apple only (condition)

Formula:-

=sumif(A1:A3, B1:B3, "Apple")

A1:A3 checks for Apple

Wherever it finds apple it adds the matching values in B1:B3

→ so it adds $40 + 20 = 60$.

* Sumif works with text, numbers and also the operators.

* It is very useful for filtering and summing based on the condition.

2) What is the difference b/w the sum and sumproduct functions?

Sum

- * It only adds numbers.
 - * Basic function.
 - * It cannot handle arrays automatically.
- Ex:- $\text{sum}(A1:A5)$
- * It works like normal calculator addition.

Ex:-

Quantity (A)

2

3

1

Price (B)

10

20

15

$\text{sum}(B1:B3)$

$\text{sumproduct}(A1:A3, B1:B3)$

↓

$$= (2 \times 10) + (3 \times 20) + (1 \times 15)$$

$$= 20 + 60 + 15$$

$$= 95$$

Sumproduct

- * It multiplies matching cells first and then adds.
- * More advanced function.
- * It works with arrays easily.
- * $\text{sumproduct}(A1:A3, B1:B3)$
- * It is used for advanced calculations like weights, averages, conditional sums etc.

3) How would you use the round, roundup and rounddown functions.

Round function :- It rounds a number to the nearest value based on normal rounding rules.

Syntax:-

=round(number, num-digits)

↓
value to be rounded

↓
how many decimals.

Ex:-

=round(12.4576, 2)

=> 12.46

Roundup function :- It always rounds the number up (towards the higher value) no matter what the decimal is.

Syntax:-

=roundup(number, num-digits)

Ex:- =roundup(12.412, 1)

Result:- 12.5 (even though lower than 5 it increases)

Rounddown :- It always rounds the number down (towards the lower value) no matter the number is high or low.

=rounddown(number, num-digits)

=rounddown(12.987, 1)

= 12.9

Round	Roundup	Rounddown
=round(11.5, 0) = 12	=roundup(12.3, 0) = 13	=rounddown(15.9, 0) = 15

4) Explain how to use the countif function to count cells that meet the conditions.
→ countif:- It counts the no. of cells that match the given condition.
syntax:-

=countif(range, criteria)

↓
group of cells.

↓
condition.

Ex:-

If cells A1 to A10 contains department names and I need 'CSE'

then

=countif(A1:A10, "CSE")

This counts CSE from the selected cells

* It works with text, numbers and operators.

* It satisfies only one condition and criteria.

* It is useful for counting specific names, marks etc.

5) What is pi function in excel.

* The purpose of pi function in excel returns the mathematical constant π .

* $\pi = 3.1415$

* Generally π is used while working with formulas involving circles, such as:

* calculating circumference

* calculating area of circle

* other formulas that use π .

Ex:-

$$=PI()$$

$$=2 \times \pi \times r$$

$$= \pi \times r^2$$

* This is generally used in circle calculations and mathematical formulas.

6) How do you apply the sin, cos and tan functions in excel?

→ The trigonometric functions like sin, cos and tan are used to calculate trigonometric values based on the angle.

* Sin :- It returns the sine of angle.

Ex:- $=\sin(\text{number})$

Ex:- using degrees :- $=\sin(\text{radians}(30))$

* cos :- It returns the cosine of angle.

Ex:- $\cos(\text{radians}(45))$

* Tan :- It gives the tangent of an angle.

Ex:-

$$= \tan(\text{radians}(45))$$

$$= 1$$

* Mainly used in engineering, physics and geometry.

7) How can naming ranges improve the readability and management of formulas?

Naming Range:-

* It is when you give a cell or group of cells a meaningful name.

Ex:- studentslist, employeelist

→ Makes formulas easier to read without naming Range

$$= \text{sum}(A1:A10)$$

with Naming Range

$$= \text{sum}(\text{sales})$$

→ Reduce errors

* You don't have to remember the cell addresses. If wrong cell addresses are selected errors happen.

* Naming Ranges avoid this problem.

8) What are dynamic named ranges and how do you create one?

A dynamic named range in excel is a named range that automatically adjusts its size when data is added or removed. Unlike static named ranges which remain fixed, dynamic named ranges expand or shrink based on the amount of data present.

Methods to create dynamic named ranges:

→ Using offset function: - The offset function creates a range by moving a certain no. of rows and columns from a starting point

Ex:-

=offset(A1, 0, 0, count(A:A), 1)

Ex:- Using Index function: - Index is more efficient and stable than offset.

=A1:Index(A:A, count(A:A))

Advantages:

- faster performance
- Non-volatile
- suitable for large datasets.

9) How do you reference cells from another worksheet in a formula.

In excel, referencing cells from another worksheet allows us to use data stored in one sheet inside another sheet. This helps in organizing data properly and creating linked, accurate, and professional workbooks.

Syntax:

sheetname! cellReference

Ex:- Referencing a single cell
=sheet2!A1

This formula takes the value from cell A1 of sheet2.

→ Referencing a range of cells.

=sum(sheet1!A1:A20)

This adds values from A1 to A20 in sheet1.

Advantages:-

- Easy to update data
- Automatic recalculation
- clean and structured worksheets

10) what is an array formula, and how does it differ from a regular formula.

An array formula in excel is a formula that can perform multiple calculations on one or more sets of values at the same time and return either a single result or multiple results unlike regular formulas, which work on one value at a time, array formulas work on groups of value simultaneously.

Types of Array Formulas

→ Single cell array formula:
Results: one result

Ex:
 $=\text{max}(A1:A10 * B1:B10)$

→ Multi-cell array formula:-

Returns multiple results

→ Dynamic array formulas: In newer excel versions, array formulas automatically spill results into adjacent cells.

Ex: $A1:A5 * B1:B5$

This will return multiple values automatically.